# R與免費線性規劃 gMOIP 套件-Part 1

## 大數據分析

- R/Python/Julia/SQL程式設計與應用 (R/Python/Julia/SQL Programming and Application)
- 資料視覺化 (Data Visualization)
- 機器學習 (Machine Learning)
- 統計品管 (Statistical Quality Control)
- 最佳化 (Optimization)

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# 大綱

- 1.R 簡介與安裝
- 2.Rtools 簡介與安裝
- 3.RStudio 簡介與安裝
- 4.gMOIP 套件簡介
- 5.gMOIP 線性規劃應用
- 6.結論
- Part2: ggplot2 套件簡介
- Part2: gMOIP + ggplot2 應用

# 1.R 簡介與安裝

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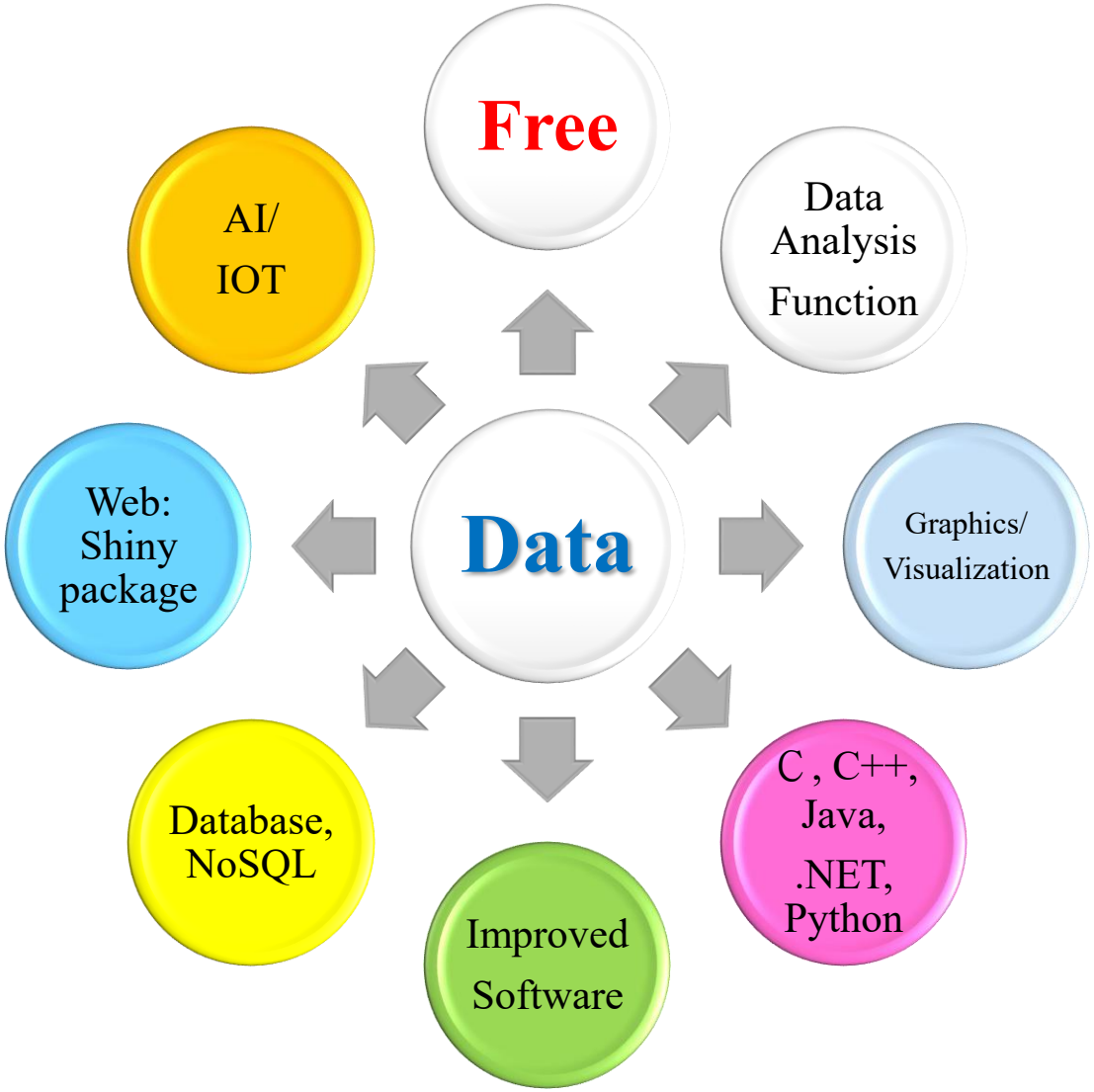
# 認識R

- 1976 - 貝爾實驗室 John Chambers, Rick Becker, and Allan Wilks 研發S語言。
- 1993 - Ross Ihaka and Robert Gentlemen, University of Auckland, New Zealand 將S語言研發為R 語言。
  - R 是一種基於 S 語言所發展出具備統計分析、繪圖與資料視覺化的程式語言。
  - R 提供現代資料科學免費實務應用工具。
- 1997年 - R的核心開發團隊 (R development core team) 成立，專責R原始碼的修改與編寫。
  - 2000年2月 - R 1.0.0
  - 2013年3月 - R 2.15.3
  - 2026年4月 - R 4.5.3

# R-八大功能


Free
Data Analysis Function
Graphics/ Visualization
C , C++, Java, .NET, Python
Improved Software
Database, NoSQL
Web: Shiny package
AI/ IOT
Data

# R官方網頁

- https://www.r-project.org/

## The R Project for Statistical Computing

[Home]

**Download**

CRAN

下載

**R Project**

About R
Logo
Contributors
What's New?
Reporting

### Getting Started

統計計算

繪圖

R is a free software environment for statistical computing and graphics. It compiles and runs on a wide variety of UNIX platforms, Windows and MacOS. To **download R**, please choose your preferred CRAN mirror.

If you have questions about R like how to download and install the software, or what the license terms are, please read our answers to frequently asked questions before you send an email.

# R 下載 (base)

- 下載最新版R: https://cloud.r-project.org/

1

**Download and Install R**

Precompiled binary distributions of the base system and contributed packages, one of these versions of R:

- Download R for Linux (Debian, Fedora/Redhat, Ubuntu)
- Download R for macOS
- Download R for Windows

2

Subdirectories:

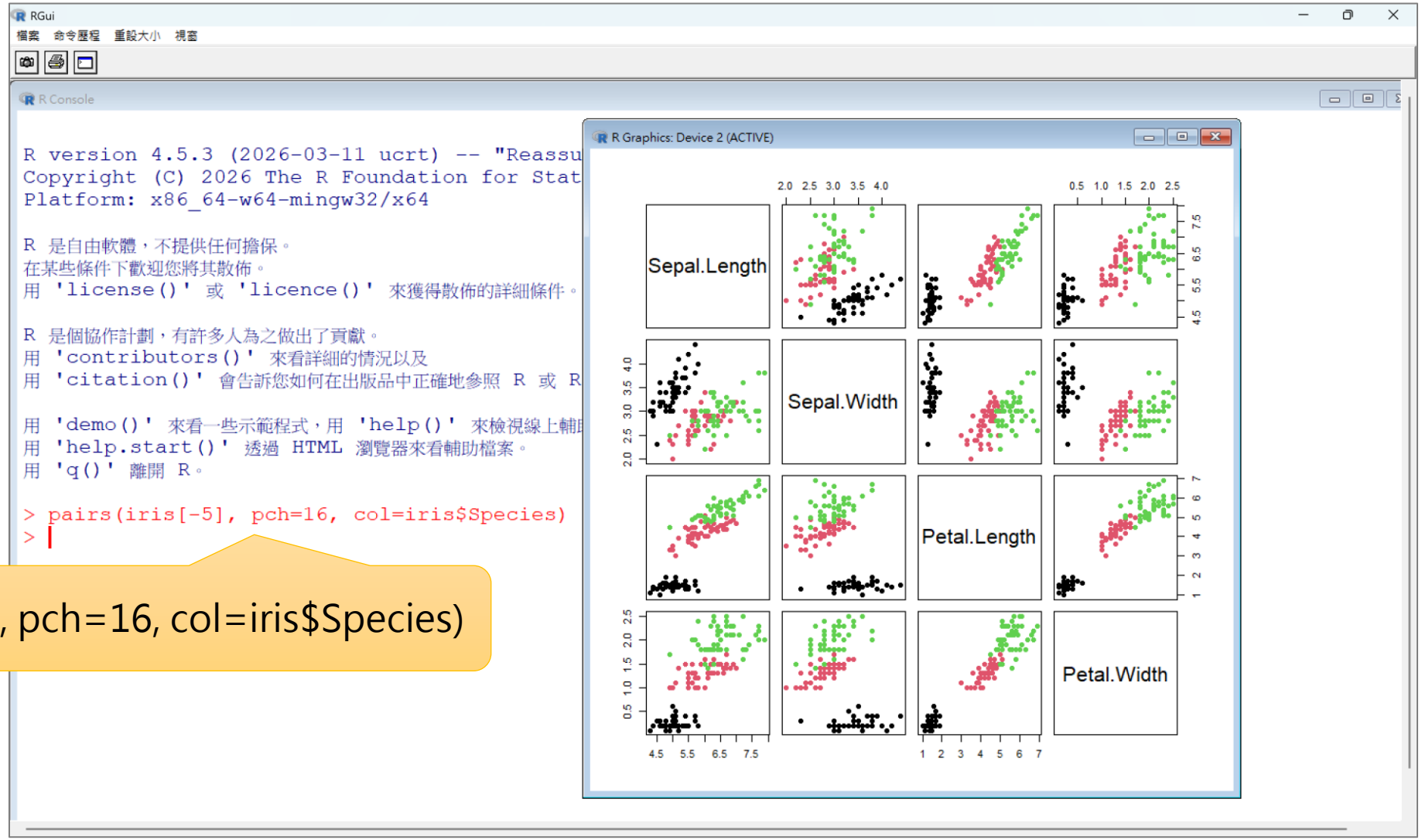
base
contrib
old contrib

Rtools

3

Download R-4.5.3 for Windows (87 megabytes, 64 bit)

# R for Windows



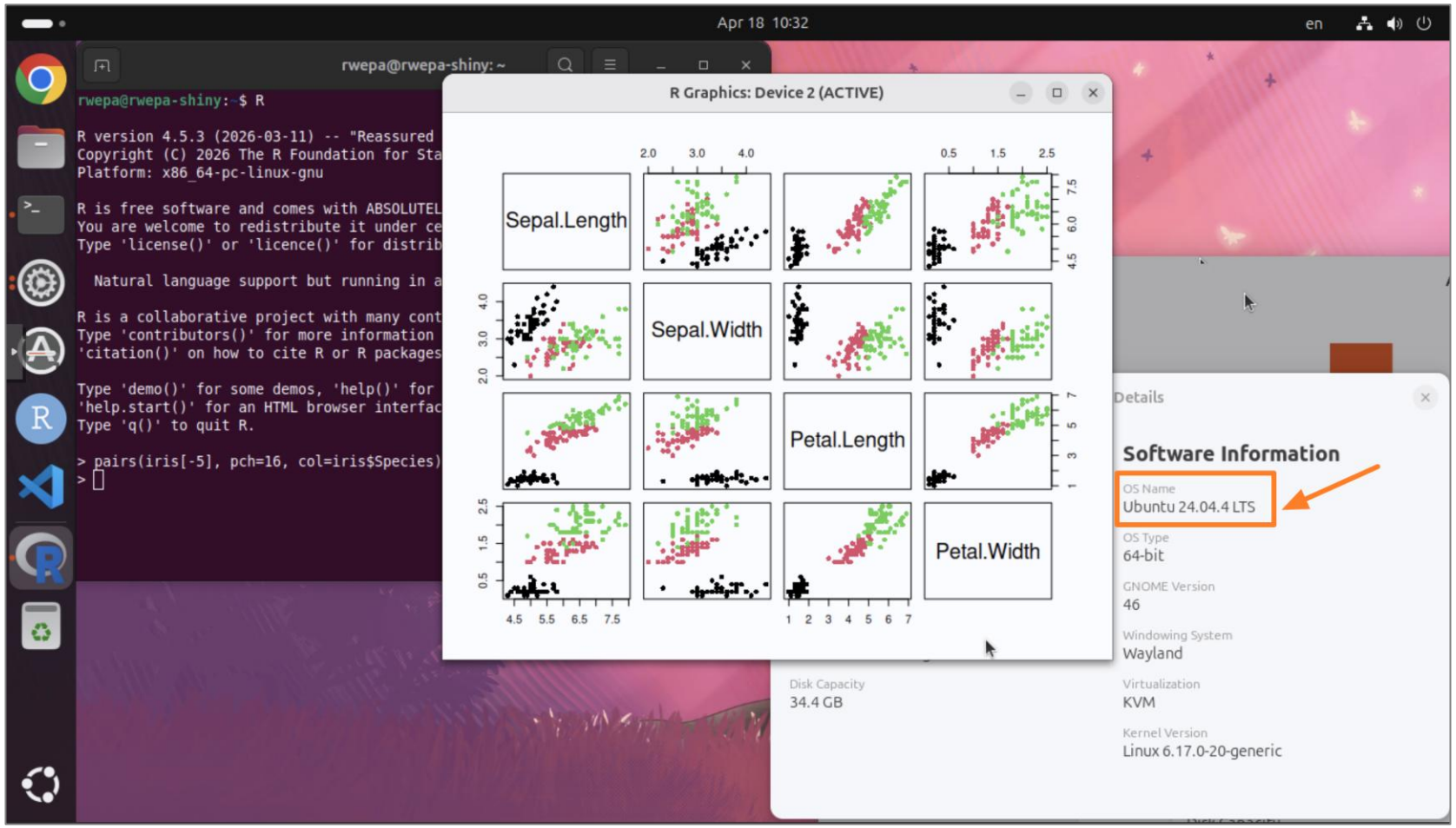

pairs(iris[-5], pch=16, col=iris$Species)

# R for macOS

R File Edit Format Workspace Packages & Data Misc Quartz Window Help
4月18日週六 上午 10:15
R Console
R version 4.5.3 (2026-03-11) -- "Reassured Reassurer"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-apple-darwin20
R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.
Natural language support but running in an English locale
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.
[R.app GUI 1.82 (8587) x86_64-apple-darwin20]
[History restored from /Users/rwepalee/.Rapp.history]
2026-04-18 10:14:21.408 R[1073:12665] TSM AdjustCapsLockLEDForKeyTrans
Inhibit
> pairs(iris[-5], pch=16, col=iris$Species)
Quartz 2 [*]
Sepal.Length
Sepal.Width
Petal.Length
Petal.Width

# R for Ubuntu

- Ubuntu R 安裝教學: https://youtu.be/DxZUj9quKEI



# 2.Rtools 簡介與安裝

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# Rtools 下載與安裝

- Rtools 是在 Windows 上使用 R 時的開發工具，主要用來編譯 R 套件（packages），特別是那些包含 C / C++ / Fortran 原始碼的套件，部分套件安裝亦須 Rtools 工具。
- Rtools for Windows: 安裝時要保留原路徑, 不要修改。
- macOS/Ubuntu 不用安裝 Rtools。

Subdirectories:

base

contrib

old contrib

Rtools

1

Choose your version of Rtools:

| | |
|---|---|
| RTools 4.5 | for R versions from 4.5.0 (R-prerelease and R-devel) |
| RTools 4.4 | for R versions 4.4.x (R-release) |
| RTools 4.3 | for R versions 4.3.x (R-oldrelease) |
| RTools 4.2 | for R versions 4.2.x |
| RTools 4.0 | for R from version 4.0.0 to 4.1.3 |
| old versions of RTools | for R versions prior to 4.0.0 |

2

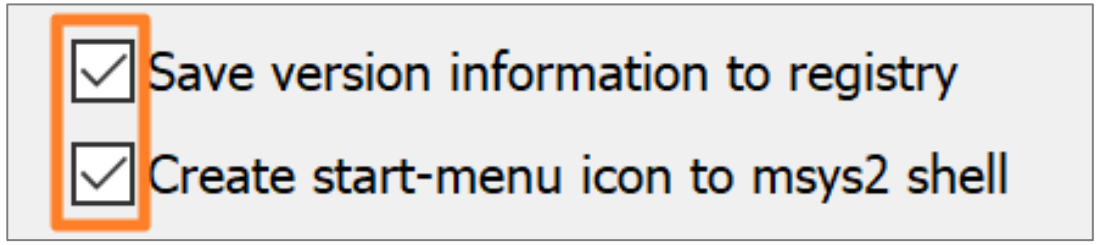
3

Rtools45 may be installed from the Rtools45 installer

- ✓ 安裝R
- ✓ 安裝 Rtools

安裝R, Windows 登入名稱:
1. 不要使用空格
2. 不要使用中文字型

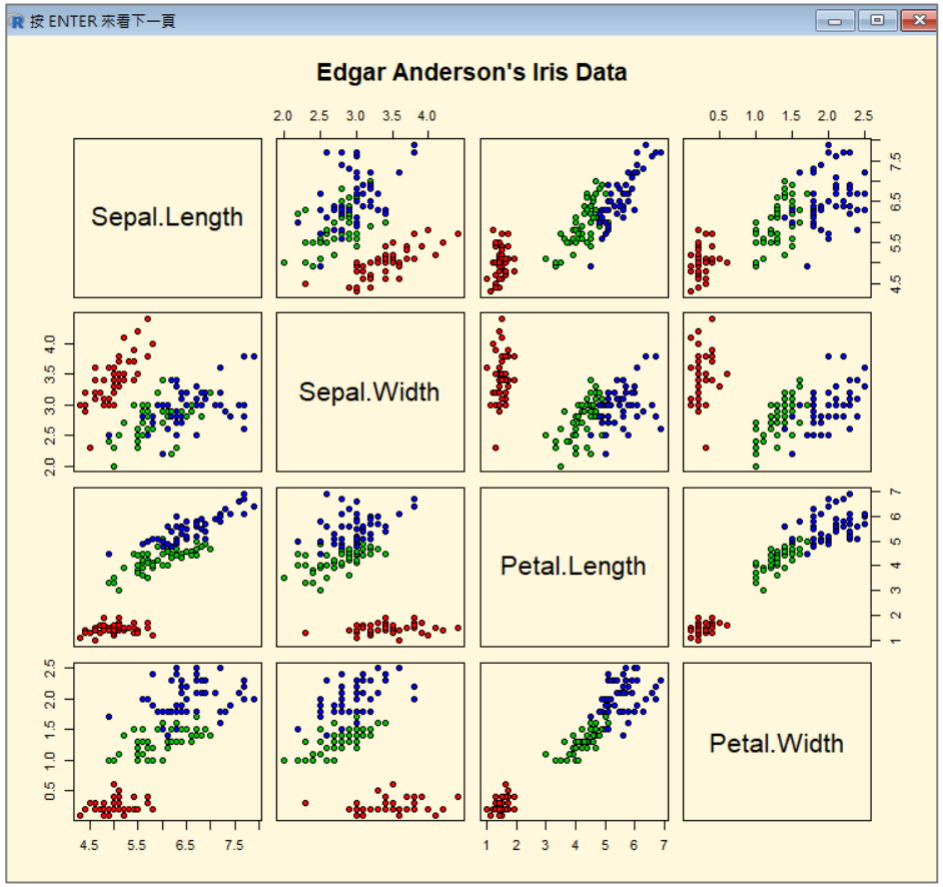
- Rtools 安裝時，二個都打勾



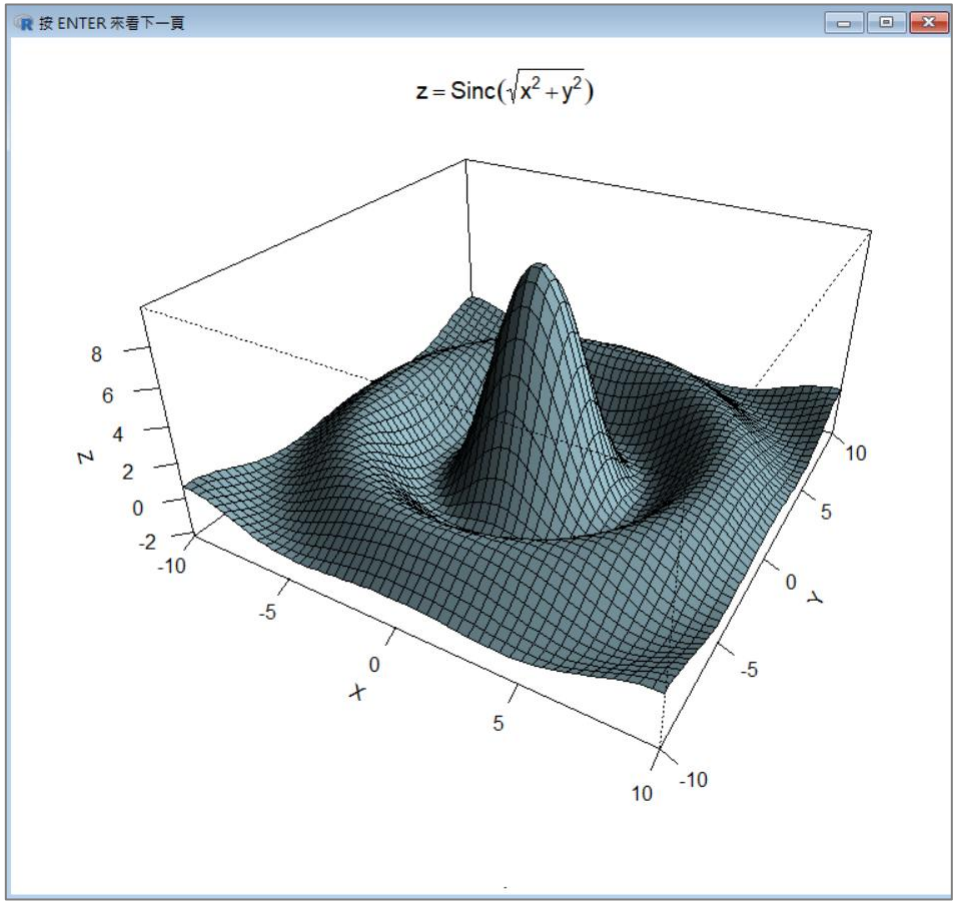

OneDrive 問題: http://rwepa.blogspot.com/2022/06/r-normalizepath-error-onedrive.html

# R 視覺化

## demo(graphics)




## demo(persp)



# 3.RStudio 簡介與安裝

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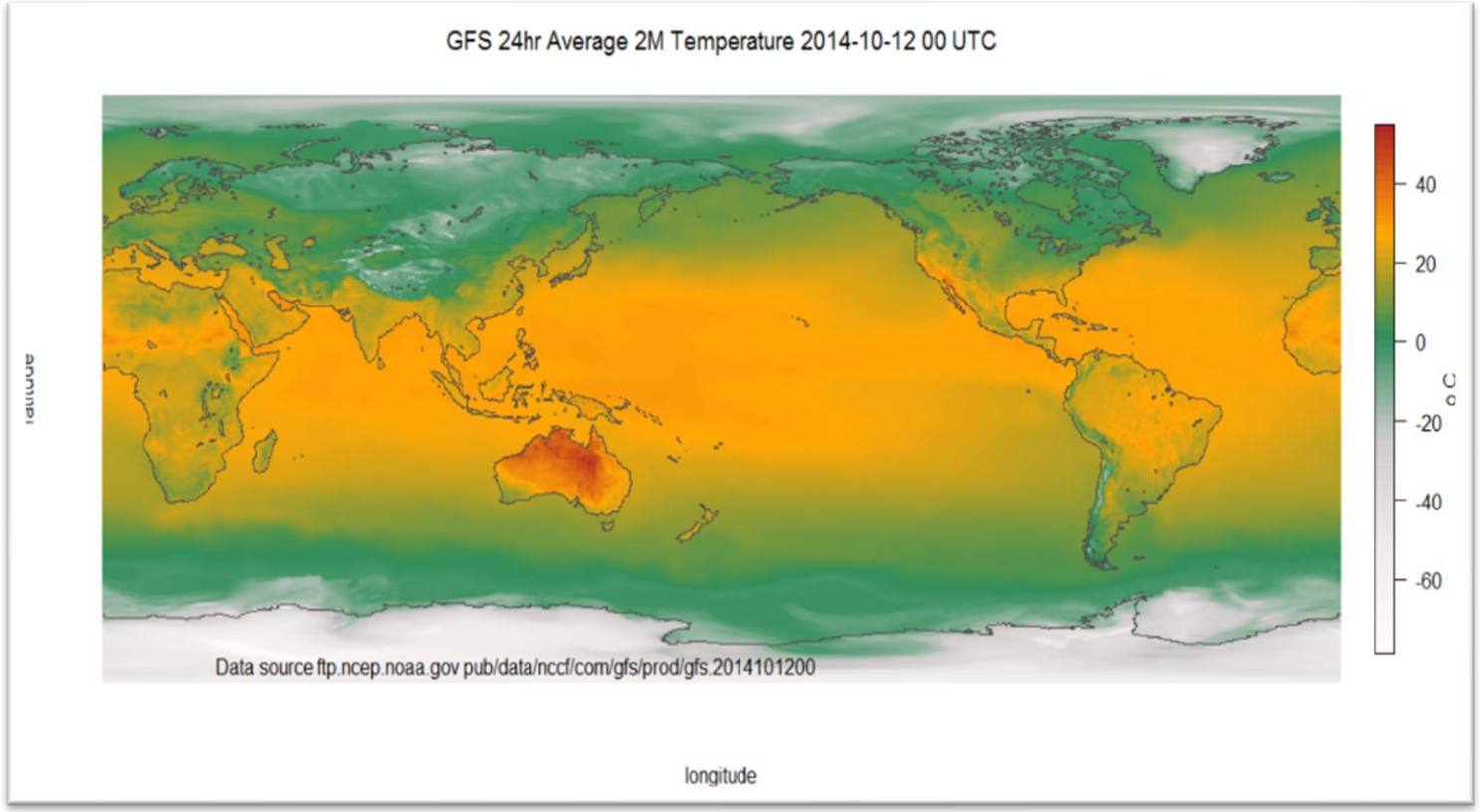
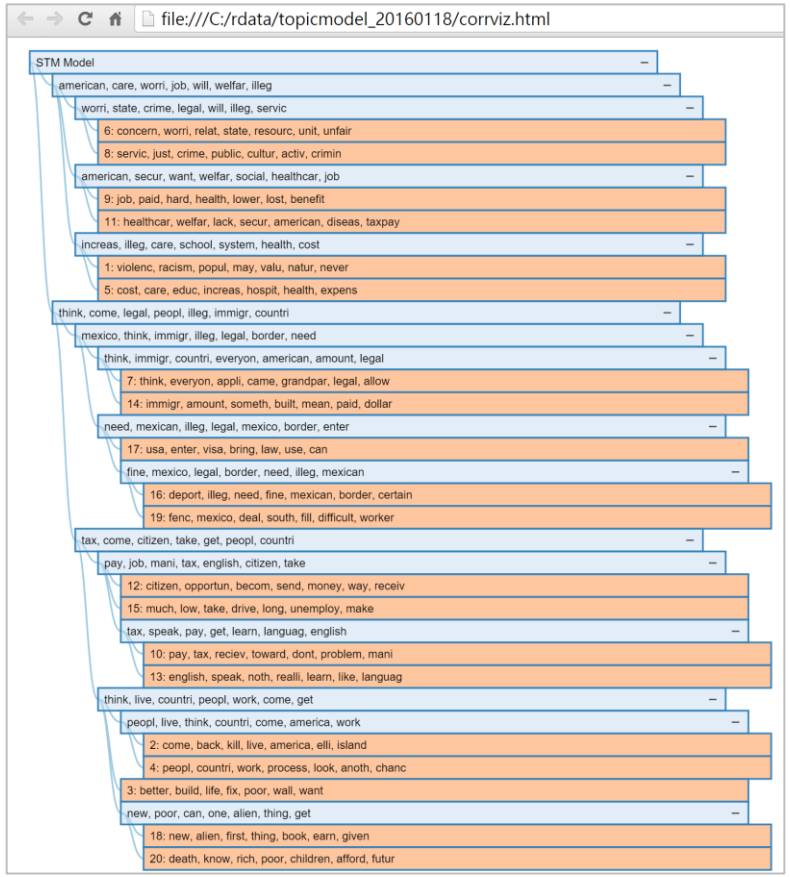
# 整合式開發環境 - RStudio




(全球2M氣溫圖)




視覺化應用

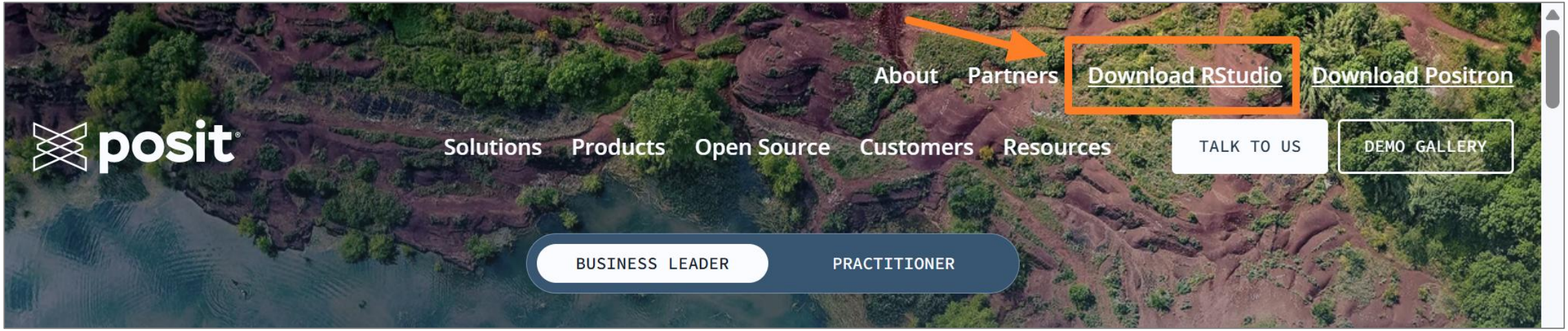
主題模型

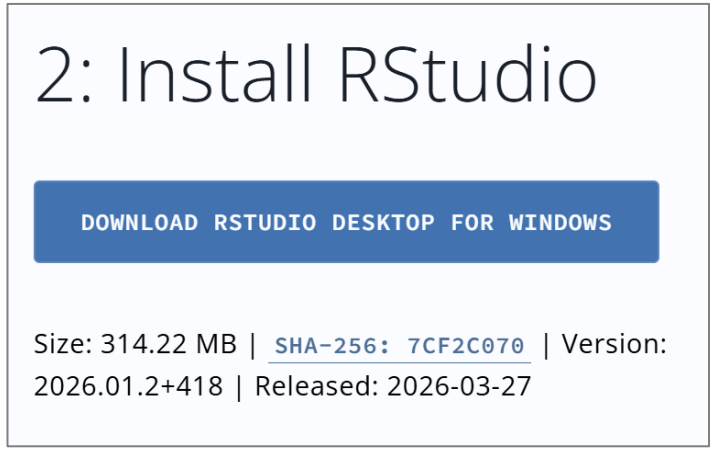
# RStudio – 特性

- 支援智慧輸入 (按Tab)
- 高亮度顯示程式碼
- 整合R/Python程式, 控制台, 變數清單, 繪圖視窗
- 整合資料庫匯入 MSSQL, MySQL, PostgreSQL, Spark…
- 整合R套件: shiny, rmarkdown, Quarto
- 支援 RStudio外掛程式 (Addins)
- 安裝注意:
  - 先安裝 R, 再安裝 RStudio
  - 安裝 RStudio時, 請先關閉R

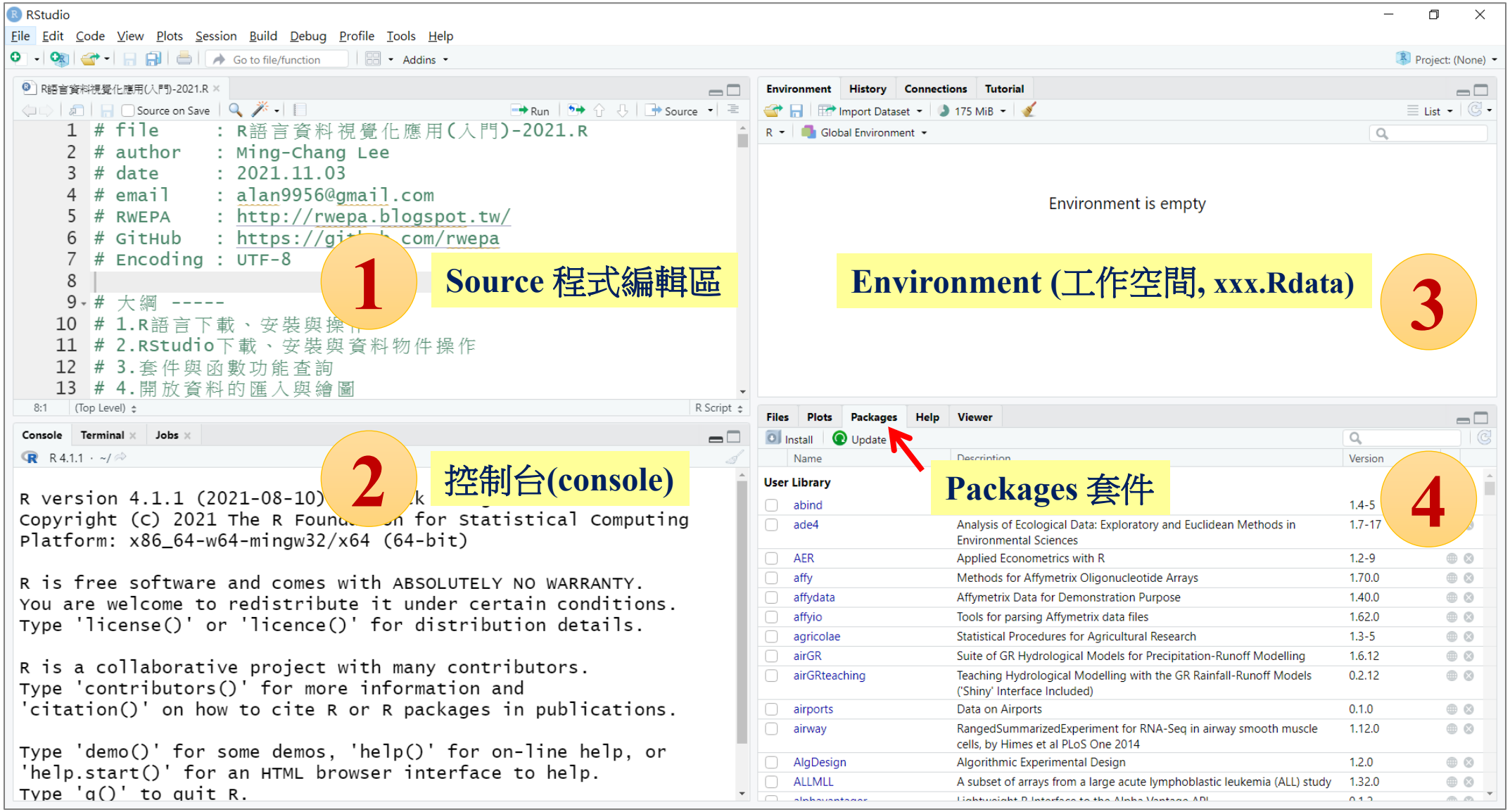
# RStudio 下載

- https://posit.co/



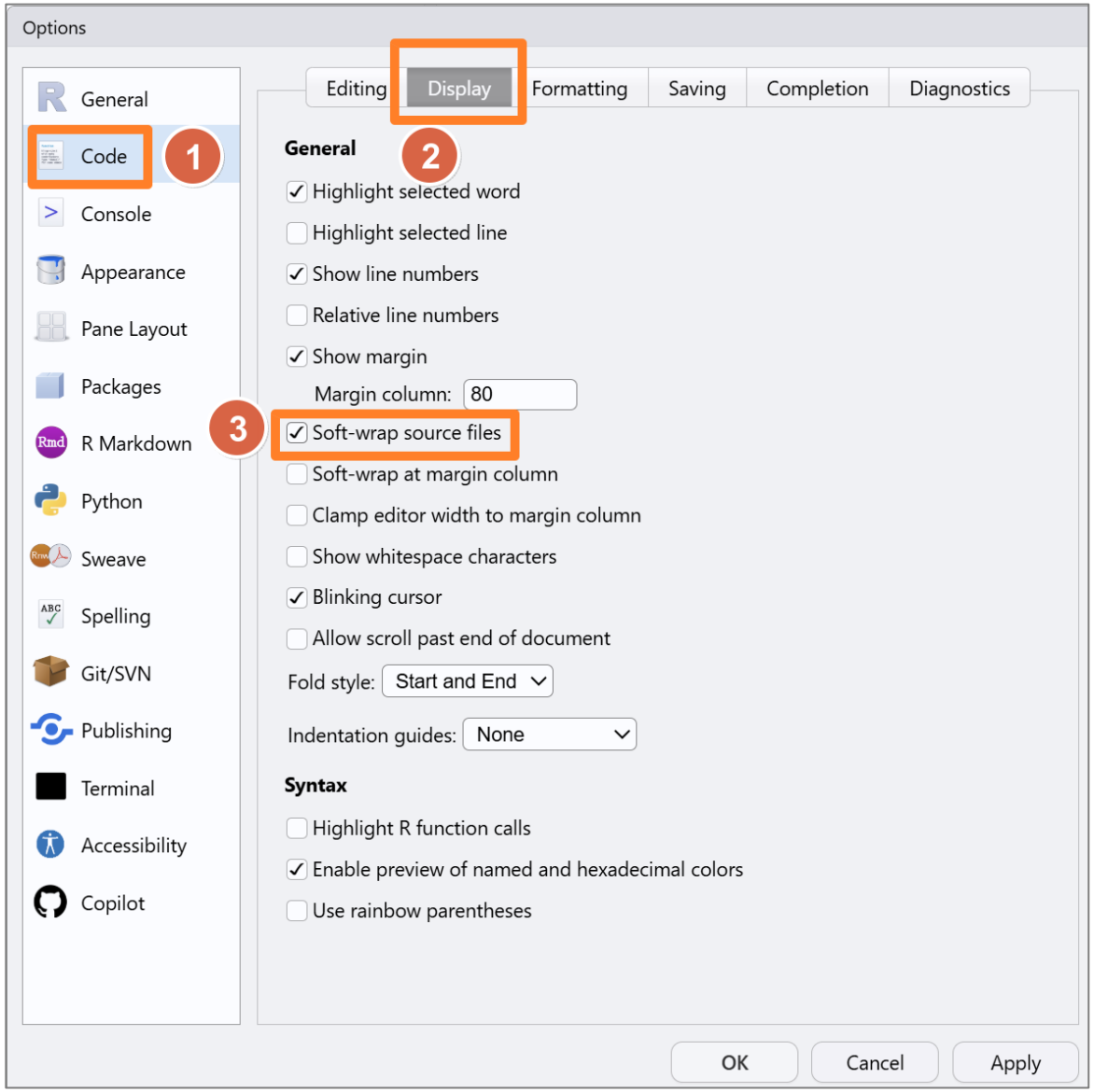



# R/RStudio環境的基礎觀念



# Soft-wrap source files (自動換列)

- Tools \ Global Options
- Code
  \ Display
  \ Soft-wrap source files 打勾



# 4.gMOIP 套件簡介

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# gMOIP 套件

- Google → r cran gMOIP
- https://cran.r-project.org/web/packages/gMOIP/index.html

**gMOIP: Tools for 2D and 3D Plots of Single and Multi-Objective Linear/Integer Programming Models**

Make 2D and 3D plots of linear programming (LP), integer linear programming (ILP), or mixed integer linear programming (MILP) models with up to three objectives. Plots of both the solution and criterion space are possible. For instance the non-dominated (Pareto) set for bi-objective LP/ILP/MILP programming models (see vignettes for an overview). The package also contains an function for checking if a point is inside the convex hull.

| | |
|---|---|
| Version: | 1.5.6 |
| Depends: | R (≥ 4.5.0) |
| Imports: | ggrepel, geometry, ggplot2, rgl, MASS, Matrix, grDevices, stats, Rfast, plyr, tidyselect, tidyr, tibble, purrr, dplyr, rlang, png, sp, moocore |
| Suggests: | tikzDevice, grid, gridExtra, knitr, rmarkdown, roxygen2, ggsci, magrittr, scales, pdftools, testthat (≥ 2.1.0), webshot2 |
| Published: | 2026-03-01 |
| DOI: | 10.32614/CRAN.package.gMOIP |
| Author: | Lars Relund Nielsen [aut, cre] |
| Maintainer: | Lars Relund Nielsen <lars at relund.dk> |
| BugReports: | https://github.com/relund/gMOIP/issues |
| License: | GPL (≥ 3) |
| URL: | https://relund.github.io/gMOIP/, https://github.com/relund/gMOIP/ |

# gMOIP 套件特色

## 1.二維/3D繪圖

- 線性規劃 (LP)
- 整數線性規劃 (ILP)
- 混合整數線性規劃 (MILP)

## 2.繪圖

- 解空間 (Solution space)
- 目標空間 (Criterion space)

## 3.二目標規劃模型(Bi-Objective LP/ILP/MILP programming models)

- 非支配解集合 non-dominated set (帕累托集合 Pareto set)

結合 ggplot2 套件

# 5.gMOIP 線性規劃應用

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# 線性規劃簡介

- 線性規劃（Linear Programming, LP）：一種在給定限制條件下，尋找最佳決策（通常是最大化利潤或最小化成本）的數學方法，廣泛應用於經濟、工程、物流與管理等領域。
- 線性規劃三大元素：
  - 決策變數（Decision Variables）：你要決定的數量，例如生產多少產品。
  - 目標函數（Objective Function）：最大化或最小化的線性函數，例如：最大化利潤，最小化成本。
  - 限制條件（Constraints）：
    描述資源或條件限制（例如原料、時間、人力），通常以線性不等式或等式表示。

# 線性規劃數學模型

- 決策變數：考慮 $n$個決策變數 $x_1, x_2, \cdots, x_n$
- 目標函數：Maximize $\{Z = c_1x_1 + c_2x_2 + \cdots + c_nx_n\}$
- 限制條件：

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &\leq b_1 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &\leq b_2 \\
&\vdots \\
x_1, x_2, \ldots, x_n &\geq 0
\end{aligned}
$$

限制式 $ax \geq b$，則左右乘以負號為 $-ax \leq -b$

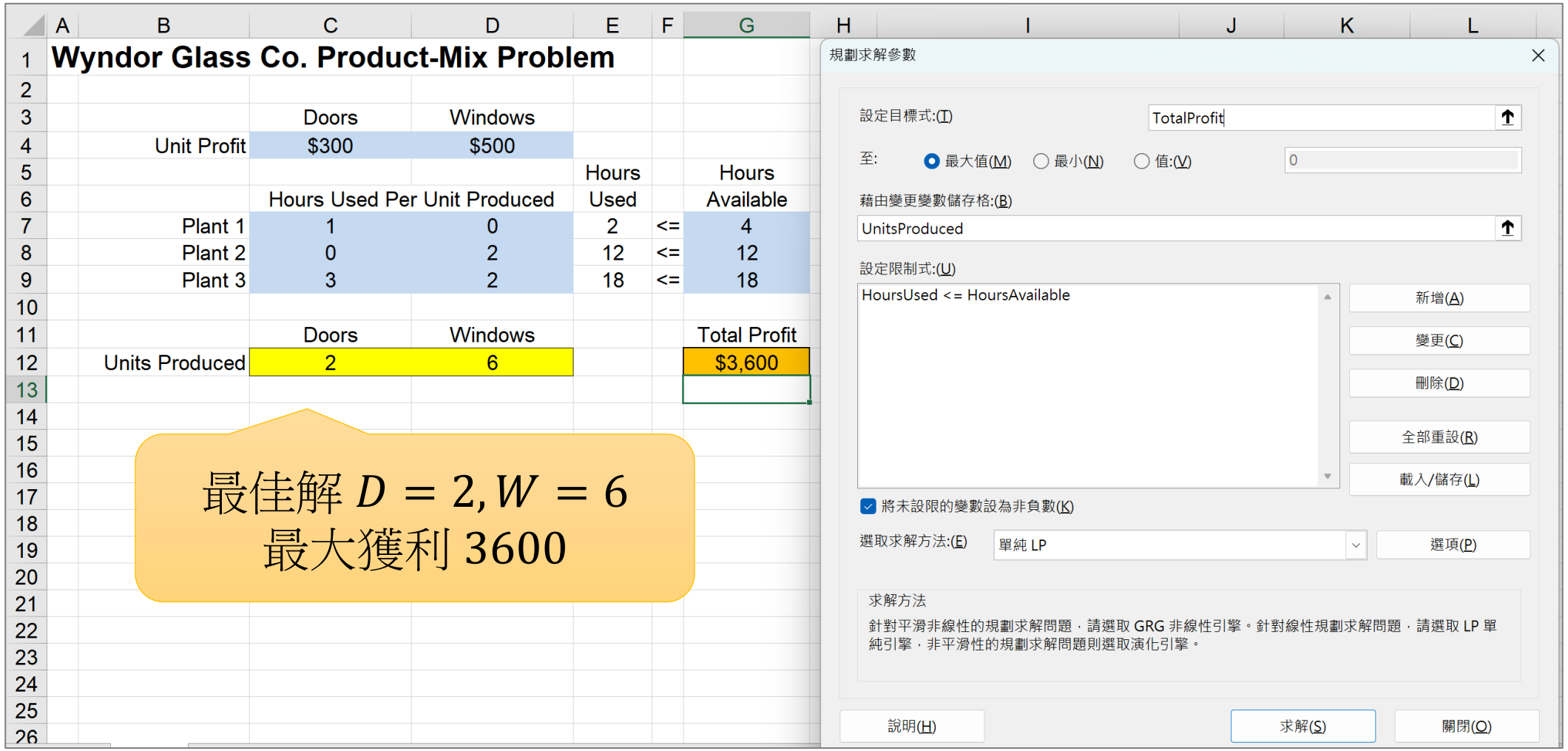
# 線性規劃最大化獲利範例

- 決策變數（Decision Variables）：
  - D = 玻璃門的生產數量
  - W = 木框窗戶的生產數量
- 目標函數（Objective Function）：
  - 利潤 $P$ 最大化 $P = 300D + 500W$
- 受限於三間工廠產能時間限制條件：

$$
\begin{aligned}
D \quad\quad\quad &\leq 4 \\
2W &\leq 12 \\
3D + 2W &\leq 18
\end{aligned}
$$

且 $D \geq 0, W \geq 0,\ D, W$*皆為整數*

# Excel 規劃求解



**Wyndor Glass Co. Product-Mix Problem**

| | Doors | Windows | Hours Used | | Hours Available |
|---|---|---|---|---|---|
| Unit Profit | $300 | $500 | | | |
| | Hours Used Per Unit Produced | | | | |
| Plant 1 | 1 | 0 | 2 | <= | 4 |
| Plant 2 | 0 | 2 | 12 | <= | 12 |
| Plant 3 | 3 | 2 | 18 | <= | 18 |

| | Doors | Windows | Total Profit |
|---|---|---|---|
| Units Produced | 2 | 6 | $3,600 |

最佳解 $D = 2, W = 6$
最大獲利 3600

規劃求解參數

設定目標式:(T) TotalProfit

至: 最大值(M) 最小(N) 值:(V) 0

藉由變更變數儲存格:(B)

UnitsProduced

設定限制式:(U)

HoursUsed <= HoursAvailable

新增(A) 變更(C) 刪除(D) 全部重設(R) 載入/儲存(L)

將未設限的變數設為非負數(K)

選取求解方法:(E) 單純 LP 選項(P)

求解方法

針對平滑非線性的規劃求解問題，請選取 GRG 非線性引擎。針對線性規劃求解問題，請選取 LP 單純引擎，非平滑性的規劃求解問題則選取演化引擎。

說明(H) 求解(S) 關閉(O)

# gMOIP 線性規劃應用

```r
# 安裝 gMOIP 套件
install.packages("gMOIP")

# R 套件安裝目錄
libPaths()
```

```
[1] "C:/Users/User/AppData/Local/R/win-library/4.5"
[2] "C:/Program Files/R/R-4.5.3/library"
```

```r
# 載入套件
library(gMOIP)
```

# 建立線性規劃模型

```
> # 建立限制條件矩陣
> A <- matrix(c(1, 0, 0, 2, 3, 2), ncol=2, byrow = TRUE)
> A
     [,1] [,2]
[1,]    1    0
[2,]    0    2
[3,]    3    2
> # 建立限制條件
> b <- c(4, 12, 18)
> b
[1]  4 12 18
> # 建立目標函數
> obj <- c(300, 500)
> obj
[1] 300 500
```

1 2 3

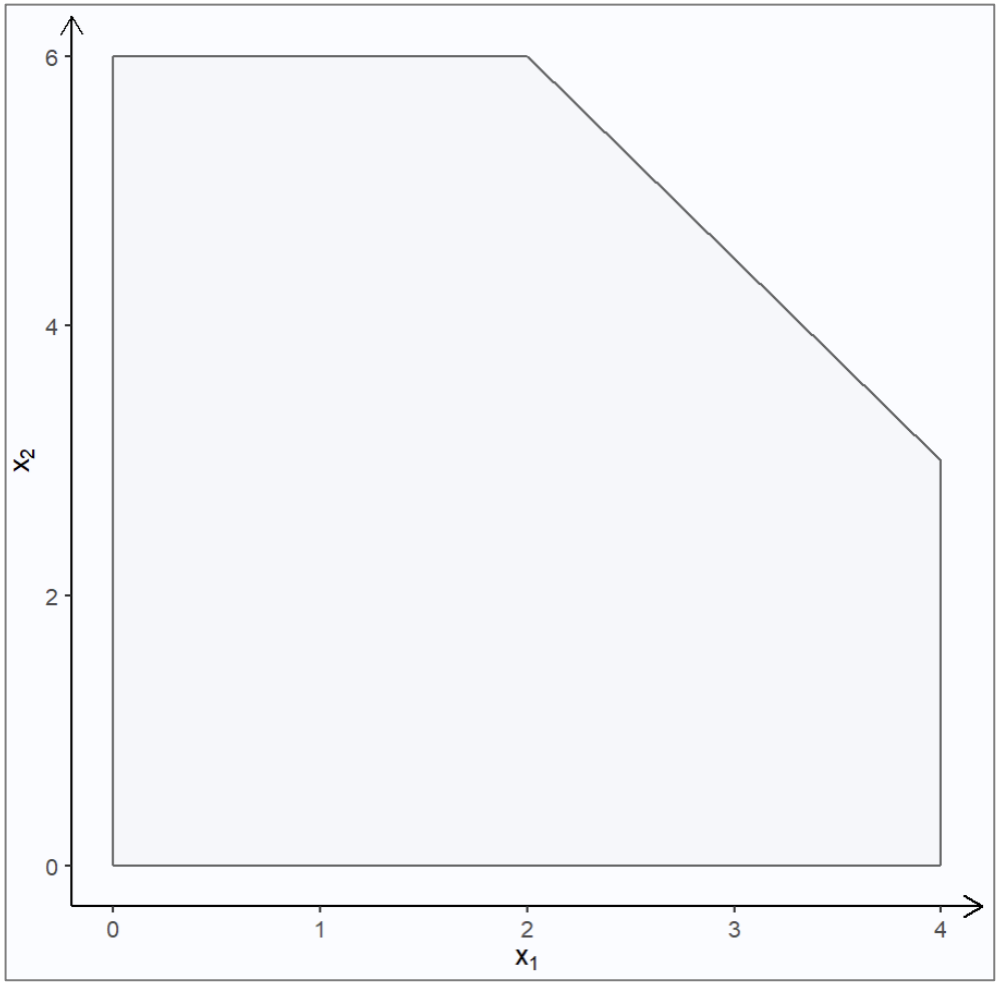
- 目標函數（Objective Function）：
  - 利潤 $P$ 最大化 $P = 300D + 500W$
- 受限於三間工廠產能時間限制條件：

$$D \leq 4$$
$$2W \leq 12$$
$$3D + 2W \leq 18$$

且 $D \geq 0, W \geq 0$, $D, W$皆為整數

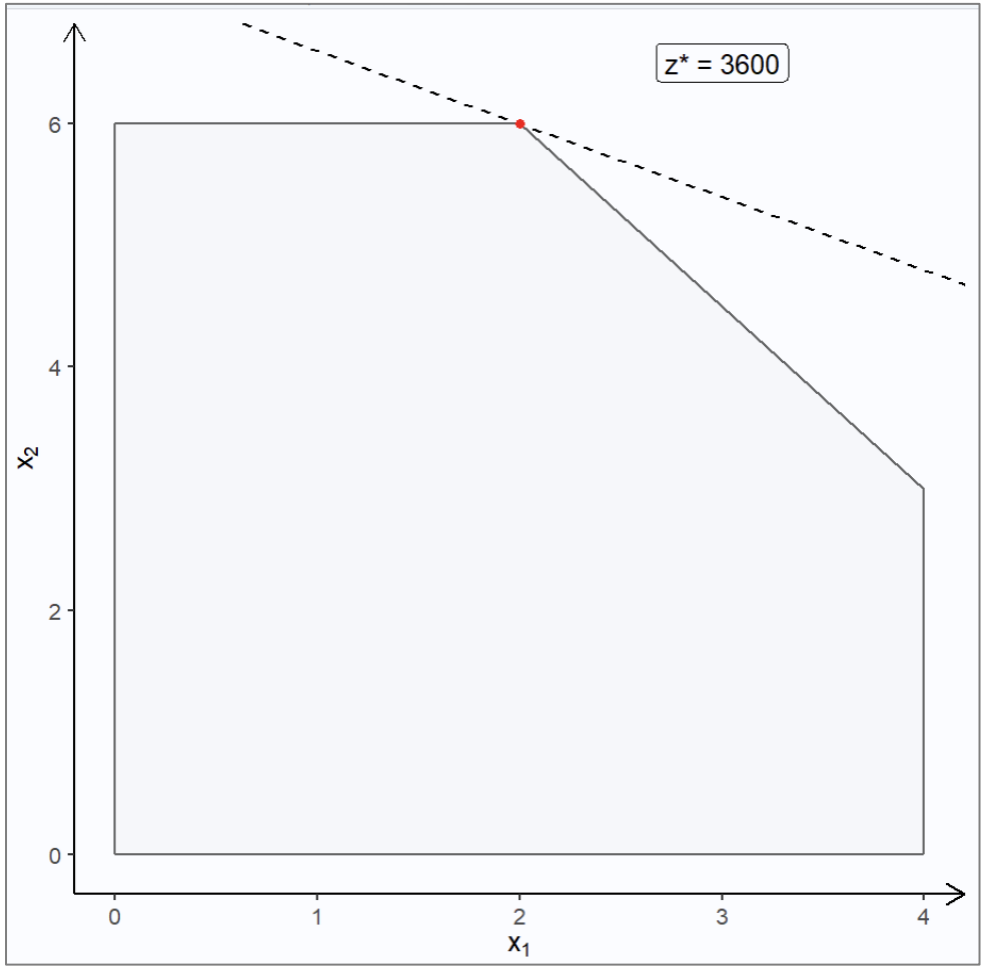
# 線性規劃圖解法

```
# 繪製凸集(Convex set)
plotPolytope(
    A,
    b,
    obj
)
```



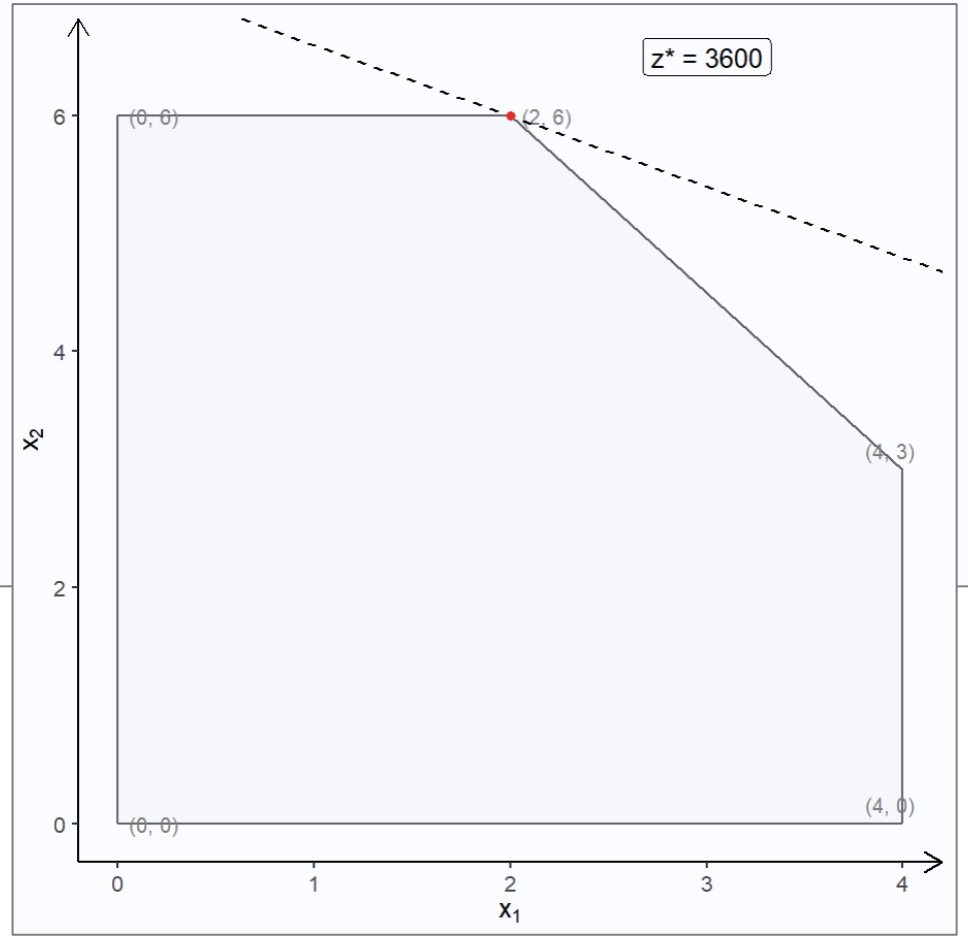
# 線性規劃圖解法+最佳解

```
# 繪製凸集(Convex set)-加入最佳值
plotPolytope(
  A,
  b,
  obj,
  plotOptimum = TRUE
)
```



# 線性規劃圖解法+最佳解+極點

```r
# 繪製凸集(Convex set)-加入最佳值,最佳解+極點(Extreme Point)
plotPolytope(
  A,
  b,
  obj,
  crit = "max",
  plotOptimum = TRUE,
  labels = "coord"
)
```



實作練習

# 練習：最小化成本範例 (1/2)

- 決策變數（Decision Variables）：
  - TV = 電視廣告的單位數量
  - PM = 平面媒體廣告的單位數量
- 目標函數（Objective Function）：
  - 成本 $C$ 最小化 $C = TV + 2PM$
- 受限滿足最小銷售量限制條件：

$$PM \geq 3$$

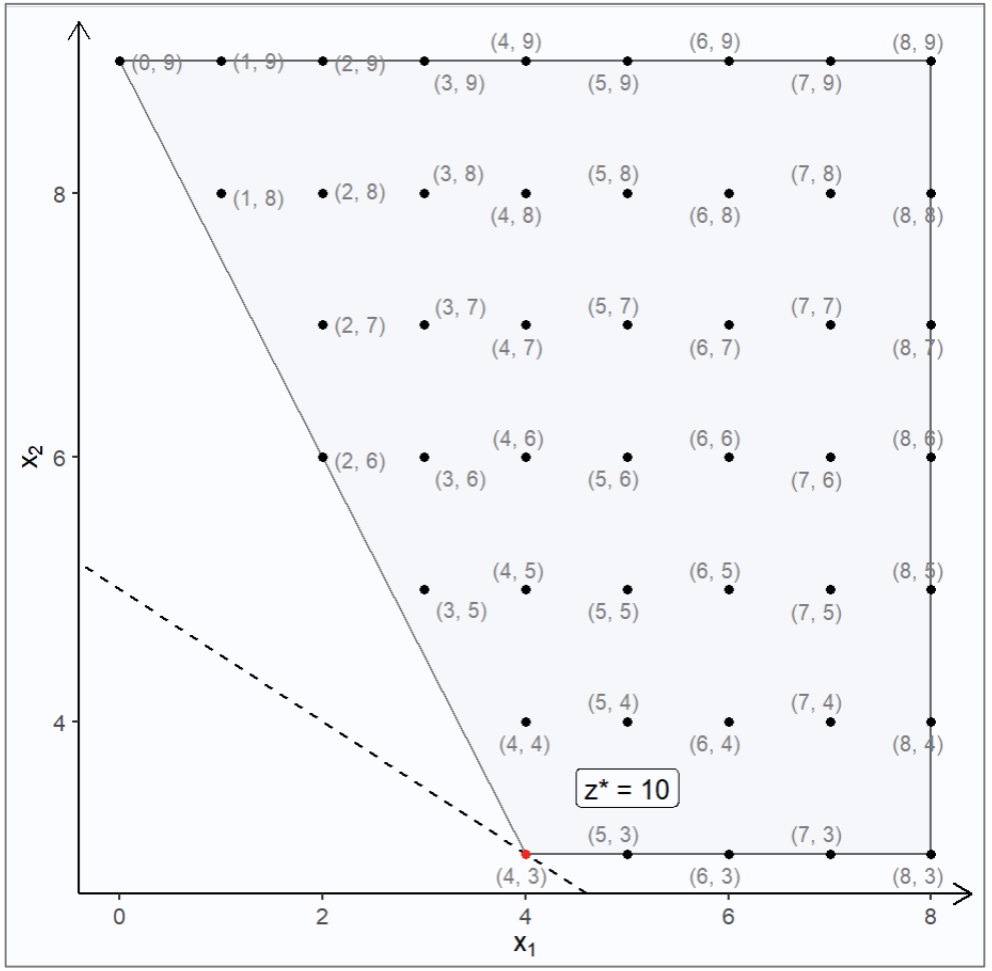
$$3TV + 2PM \geq 18$$

$$-TV + 4PM \geq 4$$

且 $\mathrm{TV} \geq 0, PM \geq 0,\ TV, PM$皆為整數

實作練習

# 練習：最小化成本範例 (2/2)



# 6.結論

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# Reviews

- 簡單說明 R 、Rtools、RStudio之功能。
- 理解 gMOIP 套件之安裝與載入。
- 熟悉最大化目標式的應用。
- 練習最小化目標式的應用。

# 參考資料

- RWEPA: http://rwepa.blogspot.com/
- YouTube: https://www.youtube.com/@alan9956
- ★★★R入門資料分析與視覺化(付費,字幕): https://mastertalks.tw/products/r?ref=MCLEE
- ★★★R商業預測與應用(付費,字幕): https://mastertalks.tw/products/r-2?ref=MCLEE
- gMOIP: https://cran.r-project.org/web/packages/gMOIP/index.html

# 謝謝您的聆聽

## Q & A

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http://rwepa.blogspot.tw/